Vahid Jebraeeli

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Summary

Ph.D. candidate specializing in generative AI and vision systems with 8+ years of experience developing scalable machine learning architectures. Research focuses on Vision Transformers, generative modeling, and data-efficient learning with publications in top-tier conferences (ICASSP, ICIP). Proven ability to design, implement, and deploy end-to-end deep learning systems from theoretical formulation through production optimization. Seeking research scientist or ML engineer roles in computer vision, generative AI, and foundational models.

Education

Ph.D. in Electrical Engineering (with Minor in Mathematics) ***2022 – Dec. 2025***
North Carolina State University, Raleigh, NC, United States
Thesis: "Balanced Scalability for Sustainable ML: Novel Data Synthesis and Transformer Dynamics for Efficient AI Systems"
Honors: Recipient of the Graduate Merit Award from the College of Engineering (2022 & 2024)
Supervisor: Dr. Hamid Krim

M.Sc. in Electrical Engineering ***2018 – 2021***
Sharif University of Technology, Tehran, Iran
Thesis: "Human identity recognition through gait and body motions analysis"
Honors: Graduated with Distinction (19.75/20 Thesis Grade); Ranked 3rd among all students in Secure Communications & Cryptography
Supervisor: Dr. Shahrokh Ghaemmaghami

B.Sc. in Electrical Engineering ***2014 – 2018***
Amirkabir University of Technology, Tehran, Iran
Thesis: "Human Identity recognition through Iris analysis by keypoint-based feature extraction method under variable image quality conditions"
Honors: Ranked in the top 0.3% (491 out of 223,000+) in the National University Entrance Exam
Supervisor: Dr. Hamidreza Amindavar

Research Interests

- Machine Learning
- Computer Vision
- Vision Language Models (vLLM)
- Agentic AI
- Model Context Protocol (MCP)
- Transformers
- Diffusion Models
- Auto-Encoders
- Generative Models
- Deep Learning
- Image and Video Analysis
- Efficient AI Systems
- 3D Computer Vision
- Multimodal Learning
- MLOps
- Responsible AI
- Large Language Models (LLMs)

Technical Skills

Programming & Core Tools:

- **Languages:** Python (Expert), MATLAB (Advanced), C++ (Intermediate)
- **Development:** Git, Docker, VS Code, PyCharm, Jupyter, Anaconda, GitHub Actions

Deep Learning & Computer Vision:

- **Frameworks:** PyTorch (Expert), PyTorch Lightning, TensorFlow/Keras, Hugging Face Transformers
- **Generative Models:** Stable Diffusion, ControlNet, VAEs, VQ-VAE, GANs, Latent Diffusion Models
- **Vision Architectures:** Vision Transformers (ViT, Swin), DETR, Deformable DETR, ConvNeXt
- **Vision-Language Models:** CLIP, BLIP-2, LLaVA, Flamingo
- **Detection & Segmentation:** MMDetection, Detectron2, YOLOv5/v8, Segment Anything (SAM)
- **3D Vision:** NeRF, 3D Gaussian Splatting, Multi-view Geometry, COLMAP, SLAM, Open3D
- **Classical CV:** SIFT, SURF, Feature Detection & Matching, Stereo Vision

Mathematical & Theoretical Foundations:

- **Optimization:** Convex Optimization, Gradient-based Methods, Adam/AdamW, Momentum-based SGD
- **Statistical Learning:** Optimal Transport Theory, Wasserstein Distance, Variational Inference
- **Advanced Theory:** Koopman Operator Theory, Volterra Systems, Expander Graphs, Manifold Learning, Dynamical Systems

Model Optimization & Deployment:

- **Inference Acceleration:** TensorRT, ONNX Runtime, OpenVINO
- **Model Compression:** Quantization (INT8/FP16), Knowledge Distillation, Pruning, LoRA
- **Edge Deployment:** TensorFlow Lite, CoreML, Mobile Optimization

MLOps & Cloud:

- **Experiment Tracking:** MLflow, Weights & Biases (W&B)
- **Cloud Platforms:** Google Vertex AI, Amazon Web Services (AWS), Microsoft Azure
- **Data Management:** Dataset Tooling, Data Versioning & Monitoring

Recent Research Projects

Volterra-Conditioned Vision Transformers — *2025–Present*

- Developing novel architecture integrating Volterra polynomial filters with Vision Transformers to capture both local non-linear features and global contextual dependencies
- Designed "Volterra-Conditioned Attention" mechanism where Volterra-generated prior tokens condition self-attention, enabling richer feature representations
- Formulating continuous-time ODE-based attention dynamics using polynomial vector fields to replace discrete Transformer blocks, improving training stability and theoretical interpretability

ViT-based Class-conditioned Autoencoder (ViTCAE) — *2025–Present*

- Architected a hierarchical Vision Transformer autoencoder (ViTCAE) that re-purposes the class token as a "generative linchpin," establishing latent dependencies that enable global semantic control over local patch-level synthesis.
- Devised a convergence-aware temperature scheduler and principled head-freezing mechanism, inspired by opinion dynamics, to reduce computational load by up to 25% FLOPs while maintaining reconstruction fidelity.
- Developed theoretically-grounded diagnostics, including an attention evolution distance and consensus functional, to adaptively control and prune attention heads during training.

Vision Transformer Training Dynamics Optimization *2024–2025*

- Investigated the evolution of attention mechanisms in Vision Transformers during training to understand and enhance model convergence and architectural efficiency
- Developed novel optimization strategies for ViT training by leveraging the Earth Mover's Distance (EMD) of attention heatmaps across iterations
- Implemented control mechanisms to guide the optimal training of ViTs through parameter evolution analysis

Expansive Synthesis for Data-Efficient Learning *2023–2024*

- Developed the "Expansive Synthesis" model to generate large-scale, information-rich datasets from minimal samples
- Utilized expander graph mappings and feature interpolation to preserve data distribution and feature relationships
- Developed a framework leveraging neural networks' non-linear latent space and Koopman operator theory
- Validated the approach by demonstrating that classifiers trained on the synthetically expanded datasets significantly outperformed those trained on the original small datasets.
- *Publication: "Generative Expansion of Small Datasets: An Expansive Graph Approach" (ICASSP 2025, 6 Citations)*

Dataset Condensation using Koopman Operator Theory *2022–2023*

- Developed the "Koopcon" framework for dataset condensation, leveraging an autoencoder architecture with Optimal Transport and Wasserstein distance objectives.
- Compressed large-scale datasets into compact representations, successfully preserving essential feature diversity, label distributions, and statistical properties.
- Demonstrated that models trained on condensed datasets achieve comparable performance to those trained on full datasets, validating significant reductions in computational and storage costs.
- *Publication: "Koopcon: A New Approach Towards Smarter and Less Complex Learning" (ICIP 2024, 7 Citations)*

Remarkable Work & Research Experiences

Graduate Research Assistant *2022–Present*
VISSTAL Lab, North Carolina State University
Advisor: Dr. Hamid Krim

- Designed and implemented novel generative architectures (ViTCAE) integrating Vision Transformers with hierarchical latent variable modeling for controllable synthesis
- Pioneered research on Vision Transformer training dynamics, developing EMD-based optimization strategies that improve convergence speed and model efficiency
- Created data synthesis framework using expander graphs and Koopman theory, enabling effective learning from minimal training samples (published ICASSP 2025)
- Developed dataset condensation methodology combining Optimal Transport theory with deep autoencoders, reducing dataset size while maintaining model performance (published ICIP 2024)

Computer Vision Intern *Summer 2023*
USDA Agricultural Research Service
Supervisor: Dr. Ebraheim Babiker

- Developed YOLOv8-based detection and instance segmentation pipeline for automated blueberry phenotyping, achieving 92% mAP on in-field data with complex backgrounds
- Implemented ML-based color calibration system for fruit maturity assessment using spectral analysis and machine learning-based ripeness classification
- Built automated measurement system for blueberry diameter estimation to support genotype classification and plant breeding research

Senior Machine Learning Engineer *2021–2022*
AI-bridge Company, Stuttgart, Germany
R&D Computer Vision Team

- Developed neural style transfer models for fashion and makeup domain transfer applications using adaptive instance normalization and feature disentanglement
- Architected robust background removal system using attention-based segmentation networks, deployed in production serving 10K+ requests/day
- Implemented text-to-image diffusion models (Stable Diffusion) for high-resolution content generation, optimizing inference pipelines for real-time synthesis

Senior Machine Learning Engineer *2020–2021*
Sharif Technology Services Complex
Quantitative Research Team

- Built cryptocurrency price prediction system using XGBoost with custom feature engineering pipeline for market microstructure data
- Designed genetic algorithm-based hyperparameter optimization framework for robust cross-validation, improving model Sharpe ratio by 35%

Graduate Research Assistant *2018–2021*
Electronic Research Institute, Sharif University
Advisor: Dr. Shahrokh Ghaemmaghami

- Developed deep learning framework for gait-based biometric identification achieving state-of-the-art accuracy on CASIA-B benchmark dataset
- Designed disentanglement architecture to separate identity features from nuisance factors (viewpoint, clothing, carrying conditions) for robust recognition
- Implemented spatiotemporal convolutional networks for motion pattern recognition invariant to viewing angles and environmental conditions

Undergraduate Research Assistant *2014–2018*
Digital Communications Lab, Amirkabir University
Advisor: Dr. Hamidreza Amindavar

- Developed iris recognition pipeline with adaptive segmentation for variable image quality and illumination conditions
- Implemented keypoint-based feature extraction using SIFT descriptors for robust biometric matching across different acquisition scenarios
- Designed score-level fusion framework combining multiple matching algorithms to improve overall recognition accuracy and system robustness

Awards & Honors

Fully Funded Ph.D. Research Assistantship *2022–2025*
North Carolina State University
Awarded full funding covering all tuition and health insurance

Graduate Merit Award (GMA) *2022 & 2024*
College of Engineering, NC State University
Recipient of prestigious graduate merit recognition

Outstanding Thesis Defense *2021 & 2018*
Defended M.Sc. & B.Sc. theses with 19.75 grade (out of 20) and best possible grade respectively

Top Rank in Specialization *2020*
Sharif University
Ranked 3rd among all students of Secure Communications & Cryptography in EE department

National M.Sc. Entrance Exam *2018*
Iran
Ranked 107 among 31,000 participants (top 0.4%) in the National Iranian Universities' Entrance Exam for Electrical Engineering M.Sc. Degree

National B.Sc. Entrance Exam *2014*
Iran
Ranked 491 among 223,000 participants (top 0.3%) in the National Iranian Universities' Entrance Exam for B.Sc. Degree

NODET Member *2010*
Tabriz, Iran
Member of National Organization for Development of Exceptional Talents

Remarkable Teaching Experiences

Summer 2020: Teaching assistant for Image Processing Course by Dr. Shahrokh Ghaemmaghami at Sharif University

Spring 2020: Teaching assistant (Head TA) for Communication Systems Course by Dr. Hamid Behrouzi at Sharif University

Spring 2020: Teaching assistant (Head TA) for Digital Communication Systems Course by Dr. Hamidreza Amindavar at Amirkabir University

Winter 2019: Teaching assistant (Head TA) for Digital Communication Systems Course by Dr. Fereydoun Behnia at Sharif University

Remarkable Academic Courses

- Deep Learning
- Advanced Machine Learning
- Digital Image Processing
- Linear Algebra
- Manifold Theory
- Speech Processing
- Random Processes
- Optimization and Algorithms
- Digital Communication Systems
- Advanced Digital Signal Processing
- Data Structure
- Queuing Systems
- Cryptography

Professional Services

IEEE ICIP: Standard Track Conference Papers Peer Reviewer. Evaluated conference papers in broad image processing and AI topics

IEEE OJSP: OJSP Track Peer Journal Papers Peer Reviewer. Assessed Open Journal of Signal Processing-track submissions in broad Signal processing and Machine Learning topics

Publications

[ICASSP 2025] Vahid Jebraeeli, Bo Jiang, Derya Cansever, and Hamid Krim. "Generative Expansion of Small Datasets: An Expansive Graph Approach", *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2025. **(6 Citations)**

[ICIP 2024] Vahid Jebraeeli, Bo Jiang, Derya Cansever, and Hamid Krim. "Koopcon: A New Approach Towards Smarter and Less Complex Learning," *IEEE International Conference on Image Processing (ICIP)*, Abu Dhabi, UAE, 2024, pp. 880-886. **(7 Citations)**

[Under Review] Vahid Jebraeeli, Hamid Krim, and Derya Cansever. "ViTCAE: ViT-based Class-conditioned Autoencoder", submitted to *ICASSP 2026*.

[In Preparation] Vahid Jebraeeli, Haoyu Yun, and Hamid Krim. "Volterra-Conditioned Attention in Vision Transformers for Enhanced Contextual Learning".